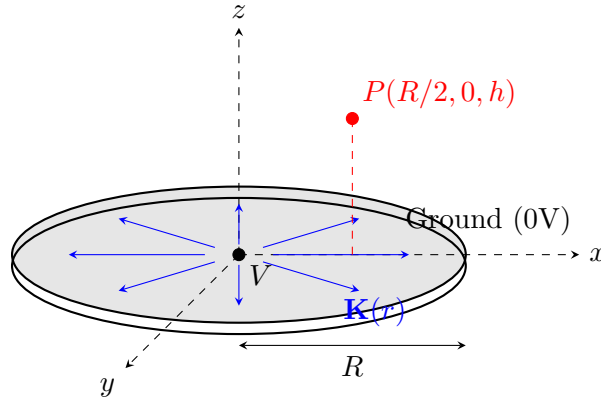


1. A thin conducting disk of radius R , thickness t ($t \ll R$), and resistivity ρ is placed in the xy -plane. The center of the disk is maintained at a potential V , while the outer circumference ($r = R$) is grounded (0V). Assume current flows uniformly and radially from the center to the edge.
 - (a) Determine the surface current density $\mathbf{K}(r)$ as a function of the distance r from the center.
 - (b) Set up the integral for the magnetic field $\mathbf{B}$ at a point $P = (\frac{R}{2}, 0, h)$, where h is the height above the disk.
 - (c) Determine which components (B_x, B_y, B_z) are non-zero at point P .



Solution

The resistance of a differential ring of radius r' and width dr' is $dR = \rho \frac{dr'}{2\pi r' t}$. Integrating from an infinitesimal inner radius r_0 to the outer radius R :

$$R_{\text{total}} = \int_{r_0}^R \frac{\rho}{2\pi t} \frac{dr'}{r'} = \frac{\rho}{2\pi t} \ln \left(\frac{R}{r_0} \right)$$

The total current is

$$I = \frac{V}{R_{\text{total}}}$$

Here we solve using surface current density (current per unit length) ($\mathbf{K}$), instead of volume current density ($\mathbf{J}$). We do this because here the thickness t is negligible compared to the radius of the disk (R) and the height of observation (h), we assume $\mathbf{J}$ is uniform across the thickness. The integral simplifies to:

$$\mathbf{K} = \mathbf{J} \cdot t = \left(\frac{I}{2\pi r t} \right) \cdot t = \frac{I}{2\pi r} \hat{r}$$

see that the thickness t cancels out. This suggests that the *net* magnetic effect at a distance is independent of how the current is distributed vertically, provided the total current I remains the same.

Using the volume density $\mathbf{J}$ requires a triple integral over r' , ϕ' , and z' :

$$\mathbf{B} = \frac{\mu_0}{4\pi} \int \int \int \frac{\mathbf{J} \times \vec{\mathcal{R}}}{\mathcal{R}^3} r' dr' d\phi' dz'$$

Substituting $\mathbf{J} = \mathbf{K}/t$ and noting that for $t \ll h$, the distance $\mathcal{R}$ is approximately constant for all z' within the plate:

$$\mathbf{B} \approx \frac{\mu_0}{4\pi} \int \int \left(\int_0^t \frac{\mathbf{K}}{t} dz' \right) \times \frac{\vec{\mathcal{R}}}{\mathcal{R}^3} r' dr' d\phi'$$

The inner integral $\int_0^t (1/t) dz'$ evaluates to 1, reducing the problem to a double integral over the surface area:

$$\mathbf{B} = \frac{\mu_0}{4\pi} \iint \frac{\mathbf{K} \times \vec{\mathcal{R}}}{\mathcal{R}^3} dA'$$

Here the surface current density $\mathbf{K}$ is the total current divided by the circumference at r' :

$$\mathbf{K}(r') = \frac{I}{2\pi r'} \hat{r}' = \frac{I}{2\pi r'} (\cos \phi' \hat{i} + \sin \phi' \hat{j})$$

We define the observation point $\mathbf{r} = d\hat{i} + h\hat{k}$ (where $d = R/2$) and source point $\mathbf{r}' = r' \cos \phi' \hat{i} + r' \sin \phi' \hat{j}$. The separation vector $\vec{\mathcal{R}} = \mathbf{r} - \mathbf{r}'$ is:

$$\vec{\mathcal{R}} = (d - r' \cos \phi') \hat{i} - r' \sin \phi' \hat{j} + h\hat{k}$$

The magnitude cubed is:

$$\mathcal{R}^3 = (d^2 + r'^2 + h^2 - 2dr' \cos \phi')^{3/2}$$

The numerator of the Biot-Savart Law $\mathbf{K} \times \vec{\mathcal{R}}$ is calculated from the determinant

$$\mathbf{K} \times \vec{\mathcal{R}} = \frac{I}{2\pi r'} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos \phi' & \sin \phi' & 0 \\ d - r' \cos \phi' & -r' \sin \phi' & h \end{vmatrix} = \frac{I}{2\pi r'} [h \sin \phi' \hat{i} - h \cos \phi' \hat{j} - d \sin \phi' \hat{k}]$$

The magnetic field is given by

$$\mathbf{B} = \frac{\mu_0}{4\pi} \int_{r_0}^R \int_0^{2\pi} \frac{\mathbf{K} \times \vec{\mathcal{R}}}{\mathcal{R}^3} r' d\phi' dr'$$

Substituting the components

$$\mathbf{B} = \frac{\mu_0 I}{8\pi^2} \int_{r_0}^R \int_0^{2\pi} \frac{h \sin \phi' \hat{i} - h \cos \phi' \hat{j} - d \sin \phi' \hat{k}}{(d^2 + r'^2 + h^2 - 2dr' \cos \phi')^{3/2}} d\phi' dr'$$

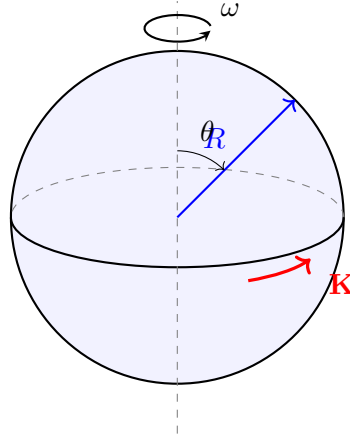
The denominator $D(\phi') = (d^2 + r'^2 + h^2 - 2dr' \cos \phi')^{3/2}$ is an **even** function of ϕ' .

- $B_x \propto \int_0^{2\pi} \frac{\sin \phi'}{D(\phi')} d\phi' = 0$ (Odd function over symmetric interval)
- $B_z \propto \int_0^{2\pi} \frac{\sin \phi'}{D(\phi')} d\phi' = 0$ (Odd function over symmetric interval)

The field is purely in the $\hat{j}$ direction. For $d = R/2$:

$$B_y = -\frac{\mu_0 I h}{8\pi^2} \int_{r_0}^R \int_0^{2\pi} \frac{\cos \phi'}{\left(\frac{R^2}{4} + r'^2 + h^2 - Rr' \cos \phi'\right)^{3/2}} d\phi' dr'$$

2. A thin, non-conducting small spherical shell of radius $R \ll r$ carries a uniform surface charge density σ . The shell rotates about the z -axis with a constant angular velocity $\omega = \omega \hat{k}$.



- Determine the expression for the surface current density $\mathbf{K}(\theta)$ at any point on the shell.
- Calculate the magnetic vector potential $\mathbf{A}(\mathbf{r})$ in all space using the integral:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \oint \frac{\mathbf{K}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} da'$$

Show that for $r > R$, it is proportional to $\frac{\sin \theta}{r^2} \hat{\phi}$.

- Also solve this using the technique mentioned in class ie

$$d\vec{A} = \frac{\mu_0}{4\pi} \frac{d\vec{m} \times \hat{r}}{r^2}$$

- Using the relation $\mathbf{B} = \nabla \times \mathbf{A}$, derive the magnetic field $\mathbf{B}$ outside the sphere.

Solution

1. Total Magnetic Dipole Moment The sphere can be treated as a collection of infinitesimal current loops (rings) at polar angle θ' with radius $a = R \sin \theta'$ and width $R d\theta'$. The current in each ring is:

$$dI = \sigma \omega R^2 \sin \theta' d\theta'$$

The magnetic dipole moment $d\mathbf{m}$ of a single ring is:

$$d\mathbf{m} = (dI)(\text{Area})\hat{k} = (\sigma \omega R^2 \sin \theta' d\theta')(\pi R^2 \sin^2 \theta')\hat{k}$$

$$d\mathbf{m} = \pi \sigma \omega R^4 \sin^3 \theta' d\theta' \hat{k}$$

Integrating from $\theta' = 0$ to π to find the total moment $\mathbf{m}$:

$$\mathbf{m} = \pi \sigma \omega R^4 \hat{k} \int_0^\pi \sin^3 \theta' d\theta'$$

Using the identity $\int_0^\pi \sin^3 x dx = \frac{4}{3}$, we find:

$$\mathbf{m} = \frac{4}{3} \pi \sigma \omega R^4 \hat{k}$$

2. Vector Potential $\mathbf{A}_{out}$ For $r \gg R$, the sphere acts as a pure magnetic dipole at the origin. The vector potential is:

$$\mathbf{A}_{out}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \hat{r}}{r^2}$$

Substituting $\mathbf{m}$ and noting that $\hat{k} \times \hat{r} = \sin \theta \hat{\phi}$:

$$\mathbf{A}_{out} = \frac{\mu_0}{4\pi r^2} \left(\frac{4}{3} \pi \sigma \omega R^4 \sin \theta \right) \hat{\phi}$$

$$\mathbf{A}_{out} = \frac{\mu_0 \sigma \omega R^4}{3} \frac{\sin \theta}{r^2} \hat{\phi}$$

Derivation of Current Density $\mathbf{K}$ from Volume Density $\mathbf{J}$

In magnetostatics, a surface current can be treated as a volume current density $\mathbf{J}$ that is confined to an infinitesimally thin shell. We define the volume current density using the Dirac delta function to constrain it to the radius R :

$$\mathbf{J}(\mathbf{r}') = \mathbf{K}(\theta') \delta(r' - R)$$

The surface current density $\mathbf{K}$ arises from the motion of the surface charge σ . For a point on the shell at (R, θ', ϕ') , the velocity due to rotation $\boldsymbol{\omega} = \omega \hat{k}$ is $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}' = \omega R \sin \theta' \hat{\phi}'$. Thus:

$$\mathbf{K} = \sigma \mathbf{v} = \sigma \omega R \sin \theta' \hat{\phi}'$$

Setting up the Vector Potential Integral

Using the above expression for $J(r')$ the vector potential $\mathbf{A}$ at an observation point $\mathbf{r}$ is given by the integral over the surface:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \oint \frac{\mathbf{K}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} da'$$

Substituting the expression for $\mathbf{K}$ and the differential area element $da' = R^2 \sin \theta' d\theta' d\phi'$:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0 \sigma \omega R^3}{4\pi} \int_0^\pi \int_0^{2\pi} \frac{\sin \theta' \hat{\phi}'}{|\mathbf{r} - \mathbf{r}'|} \sin \theta' d\phi' d\theta'$$

Evaluating the External Potential ($r \gg R$)

Step 1: Coordinate Alignment and the Vector Potential Component

We place the observation point $\mathbf{r}$ in the xz -plane, so $\phi = 0$. In spherical coordinates, $\mathbf{r} = (r, \theta, 0)$. The source point on the shell is $\mathbf{r}' = (R, \theta', \phi')$. The surface current density is $\mathbf{K} = \sigma \omega R \sin \theta' \hat{\phi}'$. Since $\hat{\phi}' = -\sin \phi' \hat{i} + \cos \phi' \hat{j}$, and our observation point is at $\phi = 0$, the local $\hat{\phi}$ direction corresponds to $\hat{j}$. The $\hat{i}$ components cancel over a full rotation (0 to 2π). Thus, we only calculate the j -component:

$$A_\phi = \frac{\mu_0 \sigma \omega R^3}{4\pi} \int_0^\pi \sin \theta' \left[\int_0^{2\pi} \frac{\sin \theta' \cos \phi'}{\sqrt{r^2 + R^2 - 2rR \cos \gamma}} d\phi' \right] d\theta'$$

where $\cos \gamma = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos \phi'$.

Step 2: Approximating the Denominator ($r \gg R$)

We factor out r from the square root and use the binomial expansion $(1 + x)^{-1/2} \approx 1 - \frac{1}{2}x$:

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{r} \left(1 + \left(\frac{R}{r} \right)^2 - \frac{2R}{r} \cos \gamma \right)^{-1/2} \approx \frac{1}{r} \left(1 + \frac{R}{r} \cos \gamma \right)$$

Substituting $\cos \gamma$:

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} \approx \frac{1}{r} + \frac{R}{r^2} (\cos \theta \cos \theta' + \sin \theta \sin \theta' \cos \phi')$$

Step 3: Performing the ϕ' Integration

We distribute the $\cos \phi'$ from the numerator into the expansion:

$$\int_0^{2\pi} \cos \phi' \left[\frac{1}{r} + \frac{R}{r^2} \cos \theta \cos \theta' + \frac{R}{r^2} \sin \theta \sin \theta' \cos \phi' \right] d\phi'$$

- Term 1: $\int_0^{2\pi} \frac{1}{r} \cos \phi' d\phi' = 0$
- Term 2: $\int_0^{2\pi} \frac{R}{r^2} \cos \theta \cos \theta' \cos \phi' d\phi' = 0$
- Term 3: $\frac{R}{r^2} \sin \theta \sin \theta' \int_0^{2\pi} \cos^2 \phi' d\phi' = \frac{R}{r^2} \sin \theta \sin \theta' (\pi)$



The ϕ' integral yields: $\frac{\pi R \sin \theta \sin \theta'}{r^2}$.

Step 4: Final Integration over θ'

Now we plug this result back into the main integral:

$$A_\phi = \frac{\mu_0 \sigma \omega R^3}{4\pi} \int_0^\pi \sin \theta' \left(\frac{\pi R \sin \theta \sin \theta'}{r^2} \right) d\theta'$$

Pulling out constants:

$$A_\phi = \frac{\mu_0 \sigma \omega R^4 \sin \theta}{4r^2} \int_0^\pi \sin^3 \theta' d\theta'$$

Using the identity $\int_0^\pi \sin^3 \theta' d\theta' = \frac{4}{3}$:

$$A_\phi = \frac{\mu_0 \sigma \omega R^4 \sin \theta}{4r^2} \left(\frac{4}{3} \right) = \frac{\mu_0 \sigma \omega R^4 \sin \theta}{3r^2}$$

Final Result

$$\mathbf{A}_{out}(r, \theta) = \frac{\mu_0 \sigma \omega R^4 \sin \theta}{3} \frac{\hat{\phi}}{r^2}$$

Calculating the Magnetic Field $\mathbf{B}$

The magnetic field is found by taking the curl of the potential, $\mathbf{B} = \nabla \times \mathbf{A}$. In spherical coordinates, for a vector with only a ϕ -component:

$$\mathbf{B} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta A_\phi) \right] \hat{r} - \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\phi) \right] \hat{\theta}$$

Substituting $A_\phi = \frac{\mu_0 \sigma \omega R^4 \sin \theta}{3r^2}$:

$$B_r = \frac{1}{r \sin \theta} \frac{\mu_0 \sigma \omega R^4}{3r^2} (2 \sin \theta \cos \theta) = \frac{2\mu_0 \sigma \omega R^4 \cos \theta}{3r^3}$$
$$B_\theta = -\frac{1}{r} \frac{\mu_0 \sigma \omega R^4 \sin \theta}{3} \left(-\frac{1}{r^2} \right) = \frac{\mu_0 \sigma \omega R^4 \sin \theta}{3r^3}$$

Final External Field Expression

$$\mathbf{B}_{out}(r, \theta) = \frac{\mu_0 \sigma \omega R^4}{3r^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta})$$

This result confirms that the spinning charged shell produces a pure dipole field in the exterior region, perfectly matching the result obtained via the magnetic moment method.

$$\mathbf{B}_{out}(r, \theta) = \frac{\mu_0 \sigma \omega R^4}{3r^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta})$$

This is the characteristic field of a perfect magnetic dipole.

3. Consider a long, straight coaxial cable of infinite length. The cable consists of:

- A solid inner conductor of radius a , carrying a total current I directed into the page.
- A thin outer cylindrical shell of radius b , carrying an equal total current I directed out of the page.

Assume the current density J in the inner conductor is uniform. Determine the magnetic field magnitude $B(r)$ as a function of the radial distance r from the central axis for the following regions:

- (a) $r < a$ (Inside the inner conductor)
- (b) $a < r < b$ (In the air gap)
- (c) $r > b$ (Outside the cable)

Now, suppose the current density in the inner conductor is not uniform, but instead varies with the distance from the axis according to the relation:

$$J(r) = Cr$$

where C is a constant. Find the magnetic field $B(r)$ for $r < a$ in terms of the total current I and the geometry of the cable.

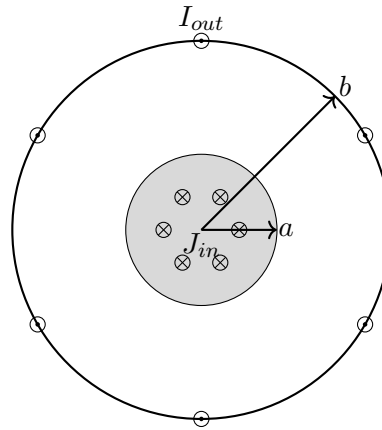


Figure 1: Cross-section of the coaxial cable.

Solution

Using **Ampere's Circuital Law**:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc}$$

Due to cylindrical symmetry, the magnetic field $\mathbf{B}$ is azimuthal (circular) and its magnitude B depends only on r . The line integral over a circle of radius r becomes:

$$B(2\pi r) = \mu_0 I_{enc}$$

Solutions for Part 1 (Uniform J)

(a) **Inside** ($r < a$): $I_{enc} = I \frac{\pi r^2}{\pi a^2} = I \frac{r^2}{a^2}$.

$$B = \frac{\mu_0 I r}{2\pi a^2}$$

(b) **Gap** ($a < r < b$): $I_{enc} = I$.

$$B = \frac{\mu_0 I}{2\pi r}$$

(c) **Outside** ($r > b$): $I_{enc} = I - I = 0$.

$$B = 0$$

Solution for Part 2 (Non-Uniform $J = Cr$) First, find C in terms of total current I :

$$I = \int_0^a J(r)(2\pi r)dr = \int_0^a (Cr)(2\pi r)dr = 2\pi C \int_0^a r^2 dr = \frac{2\pi C a^3}{3}$$
$$\Rightarrow C = \frac{3I}{2\pi a^3}$$

Enclosed current for $r < a$:

$$I_{enc}(r) = \int_0^r (Cr')(2\pi r')dr' = \frac{2\pi C r^3}{3}$$

Substitute C :

$$I_{enc}(r) = \frac{2\pi}{3} \left(\frac{3I}{2\pi a^3} \right) r^3 = I \frac{r^3}{a^3}$$

Apply Ampere's Law:

$$B(2\pi r) = \mu_0 I \frac{r^3}{a^3} \Rightarrow B = \frac{\mu_0 I r^2}{2\pi a^3}$$

4. Consider an infinitely long solenoid of radius R with n turns per unit length, carrying a steady current I . The axis of the solenoid coincides with the z -axis. Find the magnetic vector potential A in all space ($r < R$ and $r > R$). consider $\nabla \cdot A = 0$

Solution

Step 1: Derivation of the Vector Poisson Equation In magnetostatics, the two fundamental Maxwell equations are:

$$\nabla \cdot \mathbf{B} = 0 \quad (1)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad (2)$$

We can define $\mathbf{B}$ as the curl of a vector potential $\mathbf{A}$:

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (3)$$

Substituting (3) into (2):

$$\nabla \times (\nabla \times \mathbf{A}) = \mu_0 \mathbf{J} \quad (4)$$

For the solenoid, we only have a magnetic field component along $\hat{z}$, and its magnitude depends only on r , due to symmetry in z and ϕ , this gives us

$$\begin{aligned} \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} \\ \Rightarrow \frac{\partial B_z}{\partial r} \hat{\phi} &= -\mu_0 J \hat{\phi} \end{aligned} \quad (5)$$

since $B_z = \nabla \times A$

we get

$$\left(\frac{1}{r} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) \hat{r} + \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right) \hat{\phi} + \frac{1}{r} \left(\frac{\partial(rA_\phi)}{\partial r} - \frac{\partial A_r}{\partial \phi} \right) \hat{z} = B_z \hat{z}$$

once again due to symmetry, there should be no dependence on ϕ and z , so we get

$$\frac{1}{r} \frac{\partial(rA_\phi)}{\partial r} \hat{z} = B_z \hat{z} \quad (6)$$

substituting (6) into (5):

$$\frac{\partial}{\partial r} \left[\frac{1}{r} \frac{\partial(rA_\phi)}{\partial r} \right] = -\mu_0 J \quad (7)$$

Notice that we only get an equation for A_ϕ . We cannot get the value of A_r and A_z .

Since $\mathbf{B}$ has only z component, $\frac{\partial A_z}{\partial r}$ has to be 0 since it would create a B_ϕ component otherwise, so A_z has to be a constant as it can't have r dependence and can't have z or ϕ dependence. We can safely consider this to be 0 as it will have no impact and doesn't affect the form of $\mathbf{A}$.

A_r can be any function of r , as it will not contribute to any magnetic field. This makes the vector potential not unique. If we consider the coulomb gauge $\nabla \cdot \mathbf{A} = 0$, we get

$$\begin{aligned} \frac{1}{r} \frac{\partial(rA_r)}{\partial r} + \underbrace{\frac{1}{r} \frac{\partial A_\phi}{\partial \phi}}_0 + \underbrace{\frac{\partial A_z}{\partial z}}_0 &= 0 \\ \Rightarrow \frac{\partial(rA_r)}{\partial r} &= 0 \\ \Rightarrow A_r &= \frac{C}{r} \end{aligned}$$

But for this to be finite at $r = 0$, the only option is for A_r to be 0.

Step 2: Solving for the Solenoid Consider an infinite solenoid of radius R with n turns per unit length and current I . The current flows in the $\hat{\phi}$ direction. For an infinite solenoid, the current density $\mathbf{J}$ is treated as a surface current $\mathbf{K} = nI\delta(r - R)\hat{\phi}$.

This leads to:

$$\frac{d}{dr} \left[\frac{1}{r} \frac{d}{dr} (rA_\phi) \right] = 0 \quad (\text{for } r \neq R) \quad (8)$$

The general solution for both regions ($r < R$ and $r > R$) is:

$$A_\phi(r) = Cr + \frac{D}{r} \quad (9)$$

Boundary Condition: for $r < R$:

$$A(r) = C_1 r + \frac{D_1}{r}$$

but since it has to be finite at $r = 0$, we get

$$A_\phi(r) = C_1 r \quad (r < R)$$

for $r > R$:

$$A_\phi(r) = C_2 r + \frac{D_2}{r}$$

but since it has to be finite at $r = \infty$, we get

$$A_\phi(r) = \frac{D_2}{r} \quad (r > R)$$

To find the constants, we integrate the Poisson equation across the boundary from $R - \epsilon$ to $R + \epsilon$:

$$\int_{R-\epsilon}^{R+\epsilon} \frac{d}{dr} \left[\frac{1}{r} \frac{d}{dr} (r A_\phi) \right] dr = -\mu_0 \int_{R-\epsilon}^{R+\epsilon} J_\phi dr \quad (10)$$

The right side represents the surface current density $K_\phi = nI\delta(r - R)$. Thus:

$$\left[\frac{1}{r} \frac{d}{dr} (r A_\phi) \right]_{R-\epsilon}^{R+\epsilon} = -\mu_0 nI \quad (11)$$

Expanding the derivative:

$$\left(\frac{dA_\phi^{out}}{dr} + \frac{A_\phi^{out}}{r} \right) - \left(\frac{dA_\phi^{in}}{dr} + \frac{A_\phi^{in}}{r} \right) = -\mu_0 nI \quad (12)$$

Since A_ϕ must be continuous ($A_\phi^{out}(R) = A_\phi^{in}(R)$), the terms A_ϕ/r cancel, leaving:

$$\left. \frac{dA_\phi^{out}}{dr} \right|_R - \left. \frac{dA_\phi^{in}}{dr} \right|_R = -\mu_0 nI \quad (13)$$

Final Calculation

(a) **Continuity:** $C_1 R = D_2 / R \implies D_2 = C_1 R^2$.

(b) **Jump Condition:** $-\frac{D_2}{R^2} - C_1 = -\mu_0 nI \implies -C_1 - C_1 = -\mu_0 nI \implies C_1 = \frac{\mu_0 nI}{2}$.

The final result is:

$$\mathbf{A} = \begin{cases} \frac{\mu_0 nI}{2} r \hat{\phi} & r \leq R \\ \frac{\mu_0 nI R^2}{2r} \hat{\phi} & r > R \end{cases} \quad (14)$$

We can see that the magnetic vector potential is non zero everywhere, but the magnetic field of an infinite solenoid is 0 for $r > R$. So the magnetic field being 0 does not imply that the vector potential is also 0.

This question can also be solved in a much simpler way by just considering the field due to an infinite solenoid and using Stokes's theorem.

From Stokes's theorem, we get

$$\oint_C \mathbf{A} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \iint_S \mathbf{B} \cdot d\mathbf{a} = \Phi_B$$

If we consider a circular loop of radius r oriented along the z axis, we get

$$\oint \mathbf{A}_\phi \cdot d\mathbf{l} = A_\phi(r)(2\pi r)$$

A_r and A_z do not contribute to this integral. To prove that they are constant and 0 respectively, the steps are the same as that of the previous derivation, making sure $\mathbf{B}$ has only z component and from the coulomb gauge $\nabla \cdot \mathbf{A} = 0$.

For the region inside the solenoid $r < R$

Since $\mathbf{B}$ is constant $\mu_0 n I$ inside the solenoid

$$\Phi_B = \mathbf{B} \pi r^2 = \mu_0 n I \pi r^2$$

$$\begin{aligned} A_\phi(r) 2\pi r &= \mu_0 n I \pi r^2 \\ \Rightarrow A_\phi(r) &= \frac{\mu_0 n I r}{2} \end{aligned}$$

For the region outside the solenoid $r > R$

Since $\mathbf{B}$ is 0 outside the solenoid

$$\Phi_B = \mathbf{B} \pi R^2 = \mu_0 n I \pi R^2$$

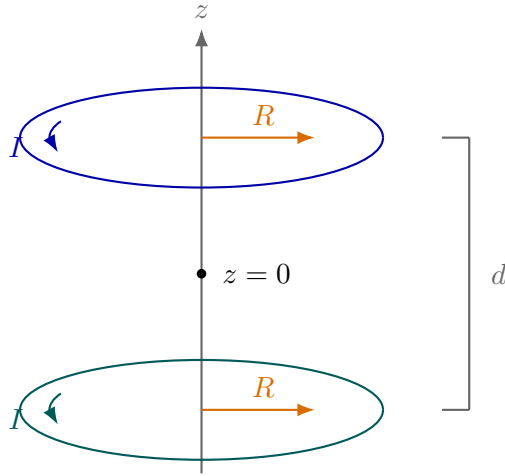
$$\begin{aligned} A_\phi(r) 2\pi r &= \mu_0 n I \pi R^2 \\ \Rightarrow A_\phi(r) &= \frac{\mu_0 n I R^2}{2r} \end{aligned}$$

$$A_\phi = \begin{cases} \frac{\mu_0 n I r}{2} & r < R \\ \frac{\mu_0 n I R^2}{2r} & r > R \end{cases}$$

5. The magnetic field on the axis of a circular current loop is given by:

$$B(z) = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}}$$

which is not uniform and falls off with increasing z . A more nearly uniform field can be produced by using two identical loops separated by a distance d along the axis.



- Find the magnetic field $B(z)$ along the axis due to the two loops, and show that $\frac{dB}{dz} = 0$ at the midpoint ($z = 0$).
- Determine the separation d such that $\frac{d^2B}{dz^2} = 0$ at the midpoint. This configuration is known as a Helmholtz coil. Also find the magnetic field at the center.
- Plot the magnetic field $B(z)$ along the axis for both $d = R$ (Helmholtz condition) and a non-optimal separation. Comment on the uniformity of the field near the center.

Solution

(a) Magnetic field on the axis due to the two loops

For one circular loop of radius R , carrying current I , the magnetic field on its axis is

$$B_{\text{loop}}(z) = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}}.$$

Now place two identical loops symmetrically about the origin, with centers at

$$z = +\frac{d}{2} \quad \text{and} \quad z = -\frac{d}{2}.$$

At a field point z on the common axis, the total field is the sum of the fields due to the two loops:

$$B(z) = \frac{\mu_0 I R^2}{2} \left[\frac{1}{\left(R^2 + \left(z - \frac{d}{2}\right)^2\right)^{3/2}} + \frac{1}{\left(R^2 + \left(z + \frac{d}{2}\right)^2\right)^{3/2}} \right]$$

and the direction is along $\hat{\mathbf{z}}$. Thus,

$$\mathbf{B}(z) = B(z) \hat{\mathbf{z}}$$

for the chosen current direction.

Show that $\frac{dB}{dz} = 0$ at the midpoint $z = 0$

Differentiate $B(z)$ with respect to z :

$$\frac{dB}{dz} = \frac{\mu_0 I R^2}{2} \left[-\frac{3(z - \frac{d}{2})}{\left(R^2 + (z - \frac{d}{2})^2\right)^{5/2}} - \frac{3(z + \frac{d}{2})}{\left(R^2 + (z + \frac{d}{2})^2\right)^{5/2}} \right] \quad (15)$$

$$= -\frac{3\mu_0 I R^2}{2} \left[\frac{z - \frac{d}{2}}{\left(R^2 + (z - \frac{d}{2})^2\right)^{5/2}} + \frac{z + \frac{d}{2}}{\left(R^2 + (z + \frac{d}{2})^2\right)^{5/2}} \right]. \quad (16)$$

Now set $z = 0$:

$$\left. \frac{dB}{dz} \right|_{z=0} = -\frac{3\mu_0 I R^2}{2} \left[\frac{-\frac{d}{2}}{\left(R^2 + \frac{d^2}{4}\right)^{5/2}} + \frac{+\frac{d}{2}}{\left(R^2 + \frac{d^2}{4}\right)^{5/2}} \right] \quad (17)$$

$$= 0. \quad (18)$$

Hence,

$$\left. \frac{dB}{dz} \right|_{z=0} = 0$$

This must happen because the arrangement is symmetric about the midpoint.

(b) Helmholtz condition: determine d such that $\frac{d^2 B}{dz^2} = 0$ at $z = 0$

To obtain a more nearly uniform field near the center, we require that not only the first derivative but also the second derivative vanish at $z = 0$.

Differentiate once more. A convenient identity is

$$\frac{d^2}{du^2}(R^2 + u^2)^{-3/2} = \frac{3(4u^2 - R^2)}{(R^2 + u^2)^{7/2}}.$$

Applying this to the two terms of $B(z)$, we get

$$\frac{d^2 B}{dz^2} = \frac{3\mu_0 I R^2}{2} \left[\frac{4 \left(z - \frac{d}{2}\right)^2 - R^2}{\left(R^2 + \left(z - \frac{d}{2}\right)^2\right)^{7/2}} + \frac{4 \left(z + \frac{d}{2}\right)^2 - R^2}{\left(R^2 + \left(z + \frac{d}{2}\right)^2\right)^{7/2}} \right].$$

Now evaluate at $z = 0$:

$$\left. \frac{d^2 B}{dz^2} \right|_{z=0} = \frac{3\mu_0 I R^2}{2} \left[\frac{4 \left(\frac{d}{2}\right)^2 - R^2}{\left(R^2 + \frac{d^2}{4}\right)^{7/2}} + \frac{4 \left(\frac{d}{2}\right)^2 - R^2}{\left(R^2 + \frac{d^2}{4}\right)^{7/2}} \right] \quad (19)$$

$$= 3\mu_0 I R^2 \frac{d^2 - R^2}{\left(R^2 + \frac{d^2}{4}\right)^{7/2}}. \quad (20)$$

For the second derivative to vanish,

$$\left. \frac{d^2 B}{dz^2} \right|_{z=0} = 0 \implies d^2 - R^2 = 0.$$

Therefore,

$$\boxed{d = R}$$

This is the **Helmholtz condition**.

Magnetic field at the center

At the midpoint $z = 0$,

$$B(0) = \frac{\mu_0 I R^2}{2} \left[\frac{1}{\left(R^2 + \frac{d^2}{4}\right)^{3/2}} + \frac{1}{\left(R^2 + \frac{d^2}{4}\right)^{3/2}} \right] \quad (21)$$

$$= \frac{\mu_0 I R^2}{\left(R^2 + \frac{d^2}{4}\right)^{3/2}}. \quad (22)$$

Under the Helmholtz condition $d = R$,

$$B(0) = \frac{\mu_0 I R^2}{\left(R^2 + \frac{R^2}{4}\right)^{3/2}} \quad (23)$$

$$= \frac{\mu_0 I R^2}{\left(\frac{5R^2}{4}\right)^{3/2}} \quad (24)$$

$$= \frac{\mu_0 I}{R} \left(\frac{4}{5}\right)^{3/2}. \quad (25)$$

Hence,

$$B_{\text{center}} = \frac{\mu_0 I}{R} \left(\frac{4}{5}\right)^{3/2}$$

and therefore

$$\mathbf{B}(0) = \frac{\mu_0 I}{R} \left(\frac{4}{5}\right)^{3/2} \hat{\mathbf{z}}$$

Key Result

For two identical circular coils of radius R , the field near the midpoint is most uniform when

$$d = R.$$

This is the **Helmholtz coil** configuration.

(c) Plot $B(z)$ for $d = R$ and for a non-optimal separation

It is convenient to normalize the field. Let

$$x = \frac{z}{R}, \quad \delta = \frac{d}{R}.$$

Then

$$B(z) = \frac{\mu_0 I}{2R} \left[\frac{1}{\left(1 + \left(x - \frac{\delta}{2}\right)^2\right)^{3/2}} + \frac{1}{\left(1 + \left(x + \frac{\delta}{2}\right)^2\right)^{3/2}} \right].$$

So the normalized field is

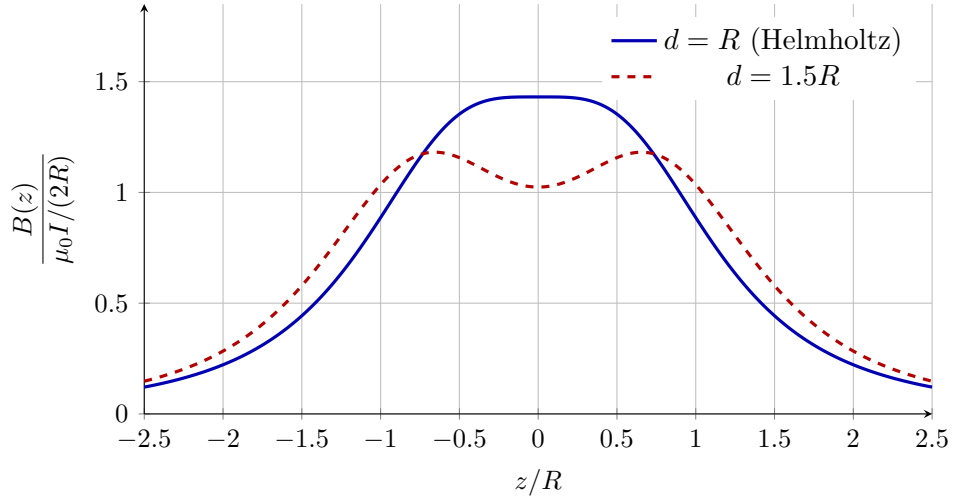
$$\frac{B(z)}{\mu_0 I / (2R)} = \frac{1}{\left(1 + \left(x - \frac{\delta}{2}\right)^2\right)^{3/2}} + \frac{1}{\left(1 + \left(x + \frac{\delta}{2}\right)^2\right)^{3/2}}$$

Below is a plot for:

$$\delta = 1 \quad (d = R, \text{ Helmholtz})$$

and

$$\delta = 1.5 \quad (d = 1.5R, \text{ non-optimal}).$$



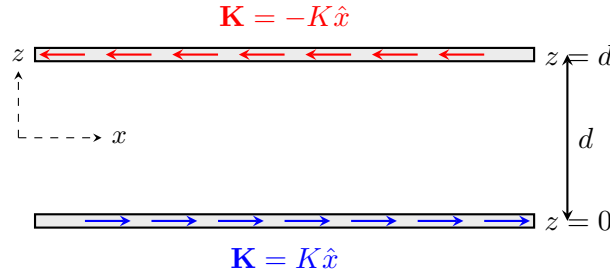
Comment on field uniformity near the center

Near $z = 0$, the Helmholtz case $d = R$ is noticeably flatter. This is because:

- symmetry always ensures $\left. \frac{dB}{dz} \right|_{z=0} = 0$,
- but only for $d = R$ do we also have $\left. \frac{d^2B}{dz^2} \right|_{z=0} = 0$.

Therefore, the field varies more slowly near the center for the Helmholtz separation.

6. An infinite plane at $z = 0$ carries a uniform surface current density $\mathbf{K} = K \hat{x}$, where K is a constant (in A/m). A second infinite plane at $z = d$ carries a surface current $\mathbf{K} = -K \hat{x}$ (equal magnitude, opposite direction).



- Using Ampère's law with a rectangular loop, find the magnetic field $\mathbf{B}$ in each of the three regions: $z < 0$, $0 < z < d$, and $z > d$. Express your answer in terms of K , μ_0 , and $\hat{y}$.
- This configuration is the magnetic analogue of a parallel-plate capacitor. Identify which region has a uniform confined field and which regions have zero field, and draw a clear parallel with the electric field of two oppositely charged infinite planes.
- The two sheets are now replaced by a single slab of finite thickness δ ($\delta \ll d$) carrying a uniform volume current density $\mathbf{J} = J_0 \hat{x}$. Show that in the limit $\delta \rightarrow 0$ with $J_0 \delta = K$ held fixed, you recover the surface current result from part (a).
- Find the magnetic vector potential $\mathbf{A}$ in all three regions for the two-sheet configuration, choosing the gauge $\nabla \cdot \mathbf{A} = 0$ and requiring $\mathbf{A}$ to be continuous at the sheets. Verify that $\mathbf{B} = \nabla \times \mathbf{A}$ reproduces part (a).

Solution

Part (a) — Magnetic field in each region

By symmetry, the field from a single infinite sheet carrying $\mathbf{K} = K \hat{x}$ must point in the $\pm \hat{y}$ direction and depend only on z . Apply Ampère's law to a rectangular loop of width ℓ in the xz -plane, straddling the sheet at $z = 0$:

$$\oint \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}} \implies 2B\ell = \mu_0 K\ell \implies B = \frac{\mu_0 K}{2}.$$

The field points in $-\hat{y}$ above and $+\hat{y}$ below (right-hand rule with $\mathbf{K} = K \hat{x}$):

$$\mathbf{B}_{\text{bottom sheet}} = \begin{cases} +\frac{\mu_0 K}{2} \hat{y}, & z < 0, \\ -\frac{\mu_0 K}{2} \hat{y}, & z > 0. \end{cases}$$

For the top sheet at $z = d$ with $\mathbf{K} = -K\hat{x}$, the signs flip:

$$\mathbf{B}_{\text{top sheet}} = \begin{cases} -\frac{\mu_0 K}{2} \hat{y}, & z < d, \\ +\frac{\mu_0 K}{2} \hat{y}, & z > d. \end{cases}$$

Superposing both sheets:

$$\mathbf{B} = \begin{cases} \mathbf{0}, & z < 0, \\ -\mu_0 K \hat{y}, & 0 < z < d, \\ \mathbf{0}, & z > d. \end{cases}$$

Part (b) — Analogy with the parallel-plate capacitor

The field is perfectly confined between the sheets and zero outside — exactly mirroring the electric field of two infinite planes carrying $\pm\sigma$:

$$\mathbf{E} = \begin{cases} 0, & \text{outside,} \\ \frac{\sigma}{\epsilon_0} \hat{n}, & \text{between the plates.} \end{cases}$$

The correspondence is $\sigma/\epsilon_0 \leftrightarrow \mu_0 K$, with $\mathbf{E}$ replaced by $\mathbf{B}$ and the normal direction replaced by $\hat{y}$ (perpendicular to both $\hat{x}$ and $\hat{z}$). In both cases the “fringe fields” outside cancel by superposition, and the interior field is uniform and independent of d .

Part (c) — Slab limit recovering the surface current

A slab of thickness δ centered at $z = 0$ with uniform volume current $\mathbf{J} = J_0\hat{x}$ encloses a current $I_{\text{enc}} = J_0 \delta \ell$ through a rectangular Ampèrian loop of width ℓ :

$$2B\ell = \mu_0 J_0 \delta \ell \implies B = \frac{\mu_0 J_0 \delta}{2}.$$

Taking $\delta \rightarrow 0$ with $J_0 \delta = K$ fixed:

$$B = \frac{\mu_0 K}{2},$$

which is exactly the result from part (a). The slab becomes a mathematical surface and $J_0 \delta \rightarrow K$ is precisely the definition of surface current density. ■

Part (d) — Magnetic vector potential

We need $\mathbf{B} = \nabla \times \mathbf{A}$ with $\mathbf{B} = B_y(z) \hat{y}$ (from part a). Since $B_y = \partial A_x / \partial z - \partial A_z / \partial x$ and we expect no x -dependence, choose $\mathbf{A} = A_x(z) \hat{x}$ so that:

$$B_y = -\frac{\partial A_x}{\partial z}.$$

Integrating in each region (with the constant of integration fixed by continuity of $\mathbf{A}$ at $z = 0$ and $z = d$, and choosing $A_x = 0$ at $z = 0$):

Region 1 ($z < 0$): $\mathbf{B} = \mathbf{0}$, so $A_x = \text{const} = 0$.

$$\mathbf{A} = \mathbf{0}.$$

Region 2 ($0 < z < d$): $B_y = -\mu_0 K$, so $-\partial A_x / \partial z = -\mu_0 K$, giving $A_x = \mu_0 K z + c_1$. Continuity at $z = 0$ requires $c_1 = 0$:

$$\mathbf{A} = \mu_0 K z \hat{x}.$$

Region 3 ($z > d$): $\mathbf{B} = \mathbf{0}$, so $A_x = \text{const}$. Continuity at $z = d$ requires $A_x = \mu_0 K d$:

$$\mathbf{A} = \mu_0 K d \hat{x}.$$

In summary:

$$\mathbf{A} = \begin{cases} 0, & z < 0, \\ \mu_0 K z \hat{x}, & 0 < z < d, \\ \mu_0 K d \hat{x}, & z > d. \end{cases}$$

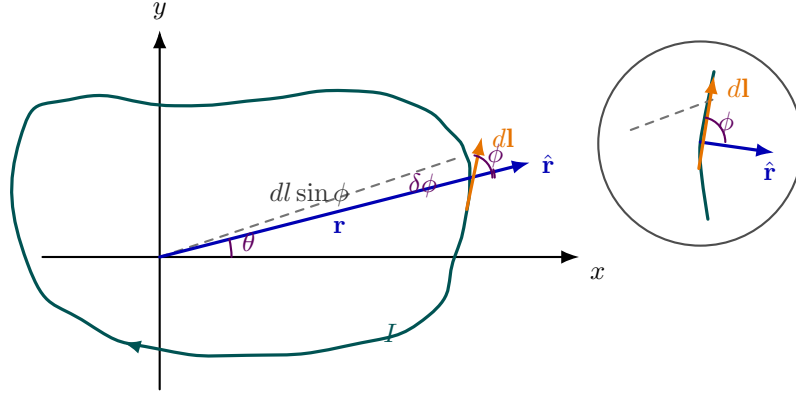
Verification: $\mathbf{B} = \nabla \times \mathbf{A}$. For $\mathbf{A} = A_x(z) \hat{x}$:

$$\nabla \times \mathbf{A} = -\frac{\partial A_x}{\partial z} \hat{y}.$$

- $z < 0$: $-\partial(0)/\partial z = 0$. ✓
- $0 < z < d$: $-\partial(\mu_0 K z)/\partial z = -\mu_0 K$, so $\mathbf{B} = -\mu_0 K \hat{y}$. ✓
- $z > d$: $-\partial(\mu_0 K d)/\partial z = 0$. ✓

All three regions match part (a). Note also that $\nabla \cdot \mathbf{A} = \partial A_x / \partial x = 0$ since A_x depends only on z , confirming the Coulomb gauge condition. ■

7. Consider a plane loop of wire that carries a steady current I ; we want to calculate the magnetic field at a point in the plane. We might as well take that point to be the origin (it could be inside or outside the loop). The shape of the wire is given, in polar coordinates, by a specified function $r(\theta)$.



- (a) Show that the magnitude of the field is

$$B = \frac{\mu_0 I}{4\pi} \oint \frac{d\theta}{r}.$$

Hint: Start with the Biot-Savart law; note that $\hat{\mathbf{z}} = -\hat{\mathbf{r}}$, and $d\mathbf{l} \times \hat{\mathbf{r}}$ points perpendicular to the plane; show that $|d\mathbf{l} \times \hat{\mathbf{r}}| = dl \sin \phi = r d\theta$.

- (b) Test this formula by calculating the field at the center of a circular loop.
(c) The “lituus spiral” is defined by

$$r(\theta) = \frac{a}{\sqrt{\theta}}, \quad (0 < \theta \leq 2\pi)$$

(for some constant a). Sketch this figure, and complete the loop with a straight segment along the x -axis. What is the magnetic field at the origin?

Solution

We place the field point at the origin O . The wire lies in the xy -plane, and its shape is given in polar form

$$r = r(\theta).$$

We will assume the current flows in the direction of increasing θ (counterclockwise). Then the magnetic field at the origin points along $+\hat{\mathbf{z}}$. If the current direction is reversed, the final answer changes sign.

(a) Show that $B = \frac{\mu_0 I}{4\pi} \oint \frac{d\theta}{r}$

Start with the Biot-Savart law:

$$d\mathbf{B} = \frac{\mu_0 I}{4\pi} \frac{d\boldsymbol{\ell} \times \mathbf{R}}{R^3},$$

where:

- $d\boldsymbol{\ell}$ is the current element,
- $\mathbf{R}$ is the vector from the current element to the field point (the origin),
- $R = |\mathbf{R}|$.

Step 1: Geometry of the source point A point on the wire has position vector

$$\mathbf{r} = r \hat{\mathbf{r}}.$$

Since the field point is the origin, the vector from the source point to the origin is

$$\mathbf{R} = -r \hat{\mathbf{r}}.$$

Hence

$$R = r.$$

Step 2: Write the line element in polar form In polar coordinates, the differential displacement along the curve is

$$d\boldsymbol{\ell} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}}.$$

Step 3: Compute the cross product Substitute $\mathbf{R} = -r\hat{\mathbf{r}}$:

$$d\boldsymbol{\ell} \times \mathbf{R} = (dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}}) \times (-r\hat{\mathbf{r}}).$$

Expand:

$$d\boldsymbol{\ell} \times \mathbf{R} = -r dr (\hat{\mathbf{r}} \times \hat{\mathbf{r}}) - r^2 d\theta (\hat{\boldsymbol{\theta}} \times \hat{\mathbf{r}}). \quad (26)$$

Now,

$$\hat{\mathbf{r}} \times \hat{\mathbf{r}} = 0, \quad \hat{\boldsymbol{\theta}} \times \hat{\mathbf{r}} = -\hat{\mathbf{z}}.$$

Therefore,

$$d\boldsymbol{\ell} \times \mathbf{R} = -r^2 d\theta (-\hat{\mathbf{z}}) = r^2 d\theta \hat{\mathbf{z}}.$$

Step 4: Substitute into Biot–Savart law Since $R = r$, we have $R^3 = r^3$. Thus

$$d\mathbf{B} = \frac{\mu_0 I}{4\pi} \frac{r^2 d\theta \hat{\mathbf{z}}}{r^3} = \frac{\mu_0 I}{4\pi} \frac{d\theta}{r} \hat{\mathbf{z}}.$$

Integrating all around the loop,

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \oint \frac{d\theta}{r} \hat{\mathbf{z}}.$$

Hence the magnitude is

$$B = \frac{\mu_0 I}{4\pi} \oint \frac{d\theta}{r}$$

and the direction is perpendicular to the plane, determined by the right-hand rule.

Equivalent geometric argument: If ϕ is the angle between $d\ell$ and $\hat{\mathbf{r}}$, then

$$|d\ell \times \hat{\mathbf{r}}| = d\ell \sin \phi.$$

From the polar-coordinate differential triangle,

$$d\ell \sin \phi = r d\theta.$$

So

$$dB = \frac{\mu_0 I}{4\pi} \frac{d\ell \sin \phi}{r^2} = \frac{\mu_0 I}{4\pi} \frac{r d\theta}{r^2} = \frac{\mu_0 I}{4\pi} \frac{d\theta}{r},$$

which is the same result.

(b) Test the formula for a circular loop

For a circle of radius R ,

$$r(\theta) = R \quad \text{for } 0 \leq \theta \leq 2\pi.$$

Then

$$B = \frac{\mu_0 I}{4\pi} \oint \frac{d\theta}{R} = \frac{\mu_0 I}{4\pi R} \int_0^{2\pi} d\theta.$$

Hence

$$B = \frac{\mu_0 I}{4\pi R} (2\pi) = \boxed{\frac{\mu_0 I}{2R}}.$$

So

$$\mathbf{B} = \frac{\mu_0 I}{2R} \hat{\mathbf{z}}$$

for counterclockwise current, which is the standard result for the magnetic field at the center of a circular loop.

(c) **Lituus spiral:** $r(\theta) = \frac{a}{\sqrt{\theta}}$, $0 < \theta \leq 2\pi$

We are given

$$r(\theta) = \frac{a}{\sqrt{\theta}}, \quad 0 < \theta \leq 2\pi.$$

Step 1: Nature of the curve As $\theta \rightarrow 0^+$,

$$r(\theta) \rightarrow \infty.$$

So the spiral starts infinitely far away along the positive x -axis.

At $\theta = 2\pi$,

$$r(2\pi) = \frac{a}{\sqrt{2\pi}},$$

so after one full turn it returns to the positive x -axis at a finite radius.

To make a closed loop, we join this endpoint back out along the x -axis by a straight segment.

Step 2: Contribution of the straight segment Along the straight segment on the x -axis, each current element $d\ell$ points along the same line as the vector joining that element to the origin. Therefore

$$d\ell \parallel \mathbf{R} \implies d\ell \times \mathbf{R} = 0.$$

Hence the straight segment contributes no magnetic field at the origin:

$$\boxed{\mathbf{B}_{\text{straight}} = 0.}$$

Step 3: Contribution of the spiral Using the result from part (a),

$$B_{\text{spiral}} = \frac{\mu_0 I}{4\pi} \int_0^{2\pi} \frac{d\theta}{r(\theta)}.$$

Now substitute $r(\theta) = a/\sqrt{\theta}$:

$$\frac{1}{r(\theta)} = \frac{\sqrt{\theta}}{a}.$$

So

$$B_{\text{spiral}} = \frac{\mu_0 I}{4\pi a} \int_0^{2\pi} \sqrt{\theta} d\theta.$$

Evaluate the integral:

$$\int \sqrt{\theta} d\theta = \int \theta^{1/2} d\theta = \frac{2}{3} \theta^{3/2}.$$

Therefore,

$$B_{\text{spiral}} = \frac{\mu_0 I}{4\pi a} \left[\frac{2}{3} \theta^{3/2} \right]_0^{2\pi} \quad (27)$$

$$= \frac{\mu_0 I}{4\pi a} \cdot \frac{2}{3} (2\pi)^{3/2}. \quad (28)$$

Simplify:

$$\frac{1}{4\pi} \cdot \frac{2}{3} (2\pi)^{3/2} = \frac{(2\pi)^{3/2}}{6\pi} = \frac{\sqrt{2\pi}}{3}.$$

Hence

$$B_{\text{spiral}} = \frac{\mu_0 I}{a} \frac{\sqrt{2\pi}}{3}$$

Since the straight segment contributes zero, the total field at the origin is

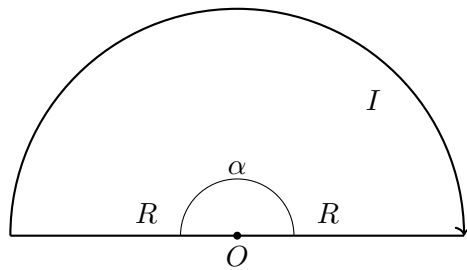
$$\mathbf{B}(0) = \frac{\mu_0 I}{a} \frac{\sqrt{2\pi}}{3} \hat{\mathbf{z}}$$

for current in the direction of increasing θ .

If the current flows in the opposite direction, then

$$\mathbf{B}(0) = -\frac{\mu_0 I}{a} \frac{\sqrt{2\pi}}{3} \hat{\mathbf{z}}.$$

8. A wire carries a steady current I and lies in the xy -plane. It consists of two parts:

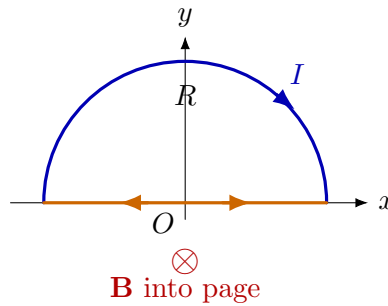


- A semicircular arc of radius R , centered at the origin, spanning from $\theta = 0$ to $\theta = \pi$.
- Two straight radial segments connecting the ends of the arc to the origin.

Thus, the wire forms a closed loop shaped like a “fan.”

- (a) Using symmetry and the Biot–Savart law, determine the direction of the magnetic field at the origin.
- (b) Show that the contribution to the magnetic field at the origin from each radial segment is zero.
- (c) Calculate the magnetic field at the origin due to the semicircular arc.
- (d) Now suppose the current in the semicircular arc flows from $\theta = 0$ to $\theta = \pi$, but the current in the radial segments flows *toward* the origin instead of away from it. Does your answer for the magnetic field at the origin change? Explain carefully.
- (e) Generalize your result: If instead of a semicircle, the arc subtends an angle α , what is the magnetic field at the origin?

Solution



(a) Direction of the magnetic field at the origin

For any current element $d\ell$ in the xy -plane, the vector from that element to the origin also lies in the xy -plane. Hence, from the Biot–Savart law,

$$d\mathbf{B} = \frac{\mu_0 I}{4\pi} \frac{d\ell \times \hat{\mathbf{r}}}{r^2},$$

the contribution $d\mathbf{B}$ must be perpendicular to the plane, i.e. along $\pm \hat{\mathbf{z}}$.

For the current direction shown in the figure, the current on the semicircular arc goes *clockwise* about the origin, so by the right-hand rule each arc element contributes a field *into the page*. Therefore the net field at the origin is along $-\hat{\mathbf{z}}$:

$$\boxed{\mathbf{B}(O) \text{ points into the page} = -\hat{\mathbf{z}}.}$$

(b) Contribution from each radial segment is zero

Consider one radial segment. At any point on that segment, the current element $d\ell$ is directed along the same straight line that joins the element to the origin. Therefore

$$d\ell \parallel \mathbf{r} \quad \implies \quad d\ell \times \hat{\mathbf{r}} = 0.$$

Hence every point on a radial segment contributes zero magnetic field at the origin. Therefore

$$\mathbf{B}_{\text{radial}}(O) = 0$$

for each radial segment, and so the total field comes entirely from the semicircular arc.

(c) Magnetic field at the origin due to the semicircular arc

For a circular arc of radius R , the distance from every current element to the origin is constant and equal to R . Also, $d\ell \perp \hat{\mathbf{r}}$, so $\sin 90^\circ = 1$. Therefore the magnitude of the Biot–Savart contribution is

$$dB = \frac{\mu_0 I}{4\pi} \frac{d\ell}{R^2}.$$

For a circular arc,

$$d\ell = R d\phi.$$

Hence

$$dB = \frac{\mu_0 I}{4\pi} \frac{R d\phi}{R^2} = \frac{\mu_0 I}{4\pi R} d\phi.$$

For a semicircle,

$$\phi : 0 \rightarrow \pi,$$

so

$$B_{\text{arc}} = \frac{\mu_0 I}{4\pi R} \int_0^\pi d\phi \quad (29)$$

$$= \frac{\mu_0 I}{4\pi R} \pi \quad (30)$$

$$= \frac{\mu_0 I}{4R}. \quad (31)$$

Since the direction is into the page for the current shown,

$$\mathbf{B}(O) = -\frac{\mu_0 I}{4R} \hat{\mathbf{z}}$$

for the loop orientation in the figure.

If only the magnitude is requested,

$$B(O) = \frac{\mu_0 I}{4R}.$$

(d) If the arc current flows from $\theta = 0$ to $\theta = \pi$, while the radial currents flow toward the origin

Now the current on the arc is *counterclockwise* as seen from $+\hat{\mathbf{z}}$. The radial pieces still lie along lines through the origin, so they still contribute zero:

$$\mathbf{B}_{\text{radial}}(O) = 0.$$

Therefore the field at the origin is again due only to the arc. Since the arc current is now *counterclockwise*, the right-hand rule says the field points *out of the page*, i.e. $+\hat{\mathbf{z}}$. Its magnitude is unchanged:

$$\mathbf{B}(O) = +\frac{\mu_0 I}{4R} \hat{\mathbf{z}}.$$

So the answer changes only in *direction*, not in magnitude.

Key point

Changing the current in the radial segments does not affect the field at the origin, because every radial segment gives zero contribution there. Only the direction of current along the circular arc matters.

(e) **Generalization: arc subtending angle α**

Now let the arc have radius R and subtend an angle α (in radians) at the center.

Exactly the same calculation applies, except the angular range is now α instead of π :

$$B_{\text{arc}} = \frac{\mu_0 I}{4\pi R} \int d\phi = \frac{\mu_0 I}{4\pi R} \alpha.$$

Thus the magnitude of the magnetic field at the origin is

$$B(O) = \frac{\mu_0 I \alpha}{4\pi R}.$$

The direction is perpendicular to the plane and is determined by the right-hand rule:

- counterclockwise current on the arc $\Rightarrow +\hat{\mathbf{z}}$ (out of page),
- clockwise current on the arc $\Rightarrow -\hat{\mathbf{z}}$ (into page).

Hence the full vector form is

$$\mathbf{B}(O) = \pm \frac{\mu_0 I \alpha}{4\pi R} \hat{\mathbf{z}},$$

where the sign is chosen according to the current direction.